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CUNEF Open Days · Workshop

What happens when companies agree to raise prices?

Antonio Alcantara, Elena Bortolato, Domenico Mergoni Cecchelli | Fall 2026

Welcome to CUNEF!

Today, you run a power company

Someone gets paid for the electricity that charges your phone.

Who decides the price?

1

The market game

You sell electricity.
Who earns the most?

Antonio

YOU ARE HERE

2

New rules

Same market,
with a twist...

Domenico

3

Data detectives

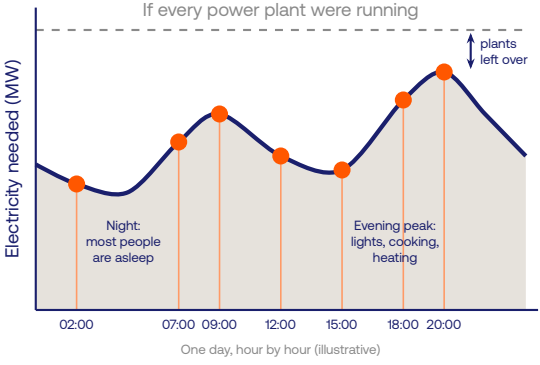
What can the numbers
tell us?

Elena

Form groups of 3–4 • one phone per group • scan the QR code on the screen

Electricity is made at the very moment it is used

So, every hour, someone must decide which power plants produce



Hard to store
It must be produced the same second it is used.

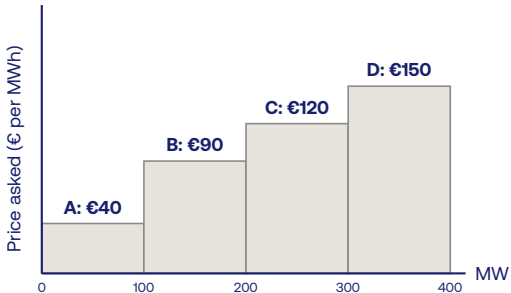
Demand changes
Low at night, highest in the evening.

Too many plants
There are always more plants than needed, so they **compete** to be chosen.

The orange dots are the 7 hours you will play.

How our market decides who produces

Cheapest offers first, until demand is covered. Each plant is paid the price it asked.



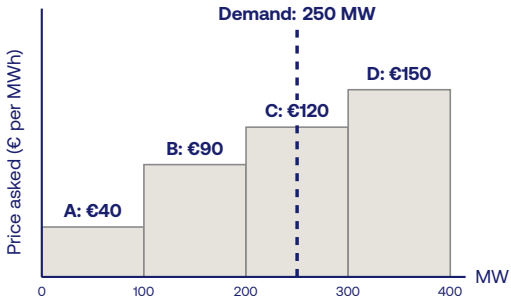
Four teams, A to D. Each has a plant of **100 MW**.

Each one sends **one price**.

The market sorts them from cheapest to most expensive.

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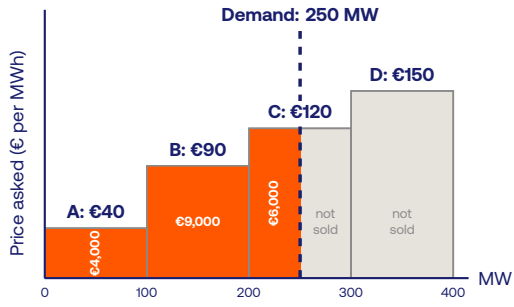


This hour, the country needs **250 MW**.

Who gets to sell?

How our market decides who produces

Cheapest offers first, until demand is covered. Each plant is paid the price it asked.



Team	Asked	Sells	Earns
A	€40	100 MW	€4,000
B	€90	100 MW	**€9,000**
C	€120	50 MW	€6,000
D	€150	0 MW	€0

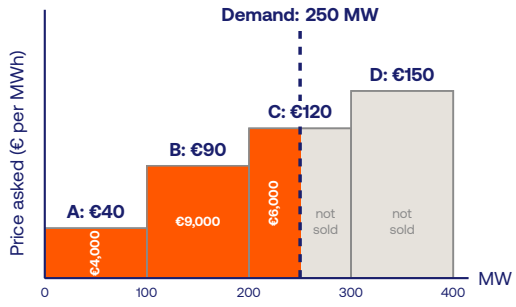
Earnings = price × MW sold (orange area).

The lowest price wins: ask too little and you earn little; ask too much and you may sell **nothing**.

So teams undercut each other: competition brings prices down.

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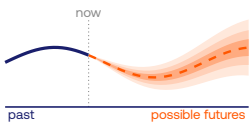
Your turn

Your plant: **100 MW**, no costs • each round is one hour • **7 rounds**
45 seconds to send one price, from €0 to €200 • the team that earns the most € wins

Optimization is everywhere, even in our game

My research: decision-making under uncertainty and uncertainty quantification.

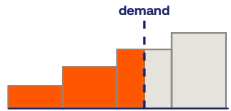
1 · Forecast



Forecast with uncertainty: demand, wind and sun

2 · Optimize

the app did this



$\min \sum_i p_i q_i$ s.t. $\sum_i q_i = D,\ 0 \le q_i \le 100$

Market clearing: the cheapest way to cover demand

3 · Bid strategically

you did this



Strategic bidding: the best price depends on the others

Professors at CUNEF work on real problems like these

and teach the maths, the data and the business behind them in three bachelor's degrees:

- Bachelor in **Data Science**
- Bachelor in **Mathematical Engineering**
- Bachelor in **Business & Technology**

Part 2 · Collusion

with Domenico Mergoni Cecchelli

Same market, same plants, same demand.

But one different strategy...

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Data detectives

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tell us?

Elena

Same teams, same phones: keep the game page open

What if the plants did *not* compete?

Agree on a high price, and everybody earns more. The customers pay for it.

Same hour, same 250 MW

	Price paid	Companies earn
They compete	€40–120	€19,000
They agree	€150	**€37,500**

Almost **double** the profit, paid by the customers.

It is as old as trade

- **1470:** Pope Paul II and King Ferdinand of Naples agreed on prices and shared the market for *alum*, the mineral that fixes dyes in Renaissance cloth.
- **Today:** Volkswagen, BMW and Daimler were fined **€875 million** by the European Commission (2021) for agreeing not to compete on emissions technology.

Agreeing is an amazing way to make money, which is exactly why it is illegal.

Play again!

Same teams, same phones, new rules

Keep the game page open.

Who earns the most this time?

Who sets the prices today?

More and more, a self-learning algorithm

The question

Each algorithm is told one thing: **earn as much money as possible**.

They **cannot communicate**.

Can they learn to collude anyway?

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My research

We study markets where the players are **learning algorithms**.

Result: independent machines, each maximising its own profit, **can learn to collude**.

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Why it matters

At CUNEF, machine learning runs through all our degrees.

Challenges ahead: **legal**, **scientific** (understanding what AI can do), and **preventive**: spotting where it happens.

Elena shows next how to detect it from data.

Part 3 · Data detectives

with Elena Bortolato

You have played 14 hours of the market.

Can the data tell us when something changed?



Everything we analyse comes from the offers you just sent

Something changed?

Consider the differences between consecutive rounds

$$R_2 - R_1,\ R_3 - R_2,\ \ldots,\ R_{15} - R_{14}.$$

Null hypothesis H_0: nothing changed

- **Fluctuations of prices are just random.**
- The order in which the prices were observed does not matter.
- We can shuffle the order of the prices, recompute the differences, and observe similar results: the same magnitude of oscillations, independent of time.

Alternative hypothesis H_1: something changed

- **Fluctuations of prices are not random.**
- The order matters in some way.
- If we shuffle the order of the prices, the pattern would be different.

Statistical test

H_0 is precise, H_1 is vague

H_0 states exactly what “nothing changed” means, while H_1 only says that there is some departure from stability, without specifying exactly *when* and *how*.

We can “recreate” the world under H_0 with simulations and compare the observed data with the artificial data generated by the simulation.

- If the data are **compatible** with the simulations of H_0, we say that there is no evidence of collusion (the information we have is not enough).
- If the data are **extreme** with respect to the simulations, then there is evidence that the order of prices matters: something has changed.

Statistical test

Or at least close enough

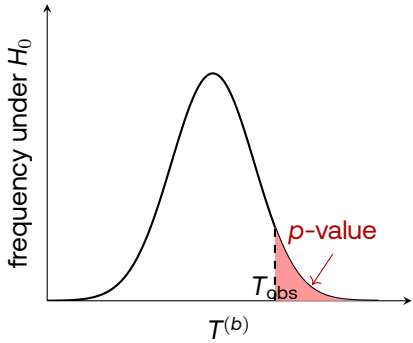
Resampling procedure

1. Choose a **test statistic** T: the largest jump, $\max_t |R_{t+1} - R_t|$.
2. Compute T_{obs} on the observed data.
3. Shuffle the prices many times (e.g. $B = 10{,}000$), recomputing $T^{(b)}$ each time. This gives the distribution of T under H_0.
4. Compute the **p-value**:

$$p = \frac{\#\{b : T^{(b)} \geq T_{\text{obs}}\} + 1}{B + 1}.$$

5. If p is small, the observed pattern would be rare under H_0: we reject H_0.

The p-value: recreating and then measuring chance



- The **p-value** is the proportion of simulated $T^{(b)}$ at least as extreme as T_{obs}.
- **Small area**: the data are extreme under H_0.
- **Large area**: the data are compatible with H_0.

Result

Decision rule

Choose a threshold $\alpha = 0.05$ and reject H_0 if the p-value is below α.

If H_0 is true, we wrongly reject it only 5% of the time (false positives).

A posteriori

If we reject H_0, we can look back at the data and identify the round responsible for the change, e.g. the round where the largest jump $|R_{t+1} - R_t|$ occurs.

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Questions?

Ask us, and
Come pick up your prizes!